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RESEARCH ARTICLE

Faster-PPENet: Advancing Logistic Intelligence for PPE Recognition at Construction Sites

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ABSTRACT The usage of Personal Protective Equipment (PPE) is crucial in a variety of sectors, including construction, manufacturing, healthcare, and hazardous material handling, to safeguard employees from possible dangers and risks. Employers may encourage a safer workplace and lower the risk of accidents and injuries by using an automated system that can determine whether people are wearing the required PPE. Several approaches have been proposed by scientists to monitor employees at workplaces, however, these techniques are limited to recognizing a small number of PPE objects. Further, historic approaches lack the generalization power to perform well in real-world cases and are not proficient in tackling several transformational alterations of the examined images like clutter, blurring, light, and brightness changes in the background settings of images. The work is focused on overcoming the limitations of existing works by proposing an effective DL strategy called the Faster-PPENet. Clearly, we have utilized the Faster-RCNN approach with a modified ResNet101 base to detect and classify various PPE. We have presented a robust feature extractor base by altering the conventional ResNet model. Specifically, we have replaced the activation approach of the ResNet approach by utilizing the Swish activation method as an alternative to the ReLU function to extract a more effective set of sample features which are later recognized by the 2-phase locator of the Faster-RCNN model. The model is evaluated on two complex databases named the Color Helmet and Vest (CHV), and Safety Helmet Detection (SHD) to show the effectiveness of our approach. Further, a real-world data sample is also utilized to show the generalization ability of our approach. For the CHV and SHD data samples, we have attained mAP scores of 0.8845, and 0.9321, along with accuracies of 90.25%, and 94.03%, while for the real-world sample, we have attained the mAP, and accuracy values of 0.901, 91% of which is clearly proving the robustness of our approach.

INDEX TERMS Deep learning, Faster-RCNN, personal protective equipment, ResNet-101, Swish, logistics Intelligence.

I. INTRODUCTION

The building sector remains among the physically demanding professions with the economy's boom. As opposed to rich nations, it requires two to ten times as many personnel per task in underdeveloped nations [1]. The building sector is also shown to be one of the riskiest among the various sectors, with a sizable number of injured workers and incidents at work. Hence, the growing size and difficulty of the building sector have led to serious difficulties in reducing the casualties at development locations. As a result, construction security guidelines and standards have been produced by organizations like the Occupational Safety and Health Administration (OSHA) in the US, the Health and Safety Executive (HSE) in the UK, and others to guarantee the security of employees [2]. Regardless of such precautions, manufacturing in the US had the greatest overall injury rate (19%) relative to other businesses in 2016–17, which is shocking [3]. Moreover, in 2018, in China, about 840 employees died at development sites. Moreover, in the UK, HSE reports that 147 personnel died from 2018 to 2019, with tumbling from a height being the most common cause of

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fatal accidents [4]. However, most accidents, injuries, and deaths might be prevented if employees wore the proper PPE, like gloves, protective glasses, safety helmets, and so forth [1]. Some of the top reasons for construction-related fatalities are falling, being electrocuted, getting entangled in machinery, colliding, and burning. By using the proper PPE, employees can avoid a lot of such incidents [5], [6]. The use of appropriate PPE on building sites is required by the applicable legislation as well as security standards. Regardless of these facts, employees frequently violate these rules because it is uncomfortable for them to put on PPE, they are unaware of potential security issues, or it impairs their work [7], [8]. As a result, it becomes crucial to track proper PPE utilization and make sure a recognition system is in place to avoid human losses. For huge places, sizable development sites, or locations with a lot of disruptions, typical PPE identification methods are, however, surprisingly ineffective. These conventional methods are subject to misclassification, false warnings, monitoring inefficiencies, high investments, and operational costs. These flaws make it impossible to track PPE effectively and send precise fire alerts [9]. As a result, scientists are now quite interested in developing automated and effective PPE recognition solutions.

The PPE tracking systems are widely classified into two types, namely the sensor-oriented or vision-based approaches. Radio Frequency Identification (RFID) tags are used in the sensor-oriented PPE monitoring method. This method involves installing RFID tags on all PPE elements. To confirm that the tags comply with the rules, they are also alternatively examined (at the entryway) or continually tracked through a Local Area Network (LAN) or wireless system [10], [11]. However, the setup and maintenance of such a large sensor system can be quite expensive, which might make it impossible to use. On the other hand, vision-based strategies examine PPE elements by capturing the environment in photographs or videos. These methods offer fairly abundant spatial data and aid in a more thorough and accurate understanding of complicated building environments [12]. These techniques also require an easy adaptive architecture. Such attributes of the vision-based approaches facilitate automated PPE recognition systems. These techniques often follow 3 major steps to develop an automated PPE identification system: i) a pre-processing phase to boost the visual appearance of the samples, ii) Keypoints computation, and iii) identification of the PPE phase, respectively. The keypoints extraction part is acknowledged as the major part of this entire procedure.

Before the introduction of convolutional neural networks (CNNs), the safety equipment detection techniques relied on the hand-coded engineered keypoints of PPE samples like safety helmet identification by employing the textural descriptors [13], edge detectors [14], and Histogram of Oriented Gradient [15] techniques, etc. along with the machine learning (ML) classifiers like random forest (RF) [16], Support Vector Machine [17] and k-Nearest Neighbor [18]). These techniques offer quick identification, but their reliability is dependent on specialist expertise. Only basic features, such as boundaries, shades, and primitive textures, may be found, even with experienced assistance. As a result, these techniques are ineffective at detecting PPE in workplaces. In addition to numerous inferential incidents in real-world scenarios, building sites frequently feature intricate PPE situations. This results in inaccurate results and inferior generalization capability of such approaches because it is very challenging to differentiate between PPEs and look-alike occurrences. The progression of artificial intelligence and ML has introduced several automated keypoints estimation approaches like CNNs, RNNs, etc., which exhibit high recognition ability in performing the classification task of several fields like healthcare diagnostics object tracing, monitoring, and autonomous vehicles, including PPE recognition. In the realm of AI and machine vision, deep learning (DL) techniques have explicitly proven to be an innovative success [19]. Prior studies used CNNs in the fields of PPE involving the identification of security safety barriers [20], elements on roofing building sites [21], employees' possibly hazardous conduct [19], typical construction-related items [22], and fire [23]. This is due to the revolutionary successes of DL approaches in accomplishing complicated tasks such as identifying objects [24]. These techniques typically use a one-phase (SSD, YOLO) [25] or a two-phase [26] (CNN, R-CNN, and Faster R-CNN) [27] recognition strategy.

Despite all these advancements, immediate PPE technologies are insufficient although they are crucial for safety. Several researchers attempted to propose PPE recognition approaches; however, still several open challenges in this field demand an appropriate approach. Existing approaches perform well in identifying one type of PPE like helmets, safety jackets, etc., however, they behave poorly in locating all of them by a single framework. Moreover, the extreme variations in the color and size of PPE also increase the complexity of the detection procedure. Further, existing works are unable to generalize well to real-world cases and unable to deal with several transformational changes of the examined images like clutter, blurring, light, and brightness changes in the background settings of images. In light of these challenges, the central research question guiding this study is: ''Can a modified Faster-RCNN model with a Swish-activated ResNet101 backbone improve multi-class PPE detection performance and generalization in real-world conditions?'' We hypothesize that replacing the ReLU activation function with Swish in the ResNet101 backbone will enhance feature extraction, resulting in improved accuracy and robustness across diverse datasets and real-world environments. Therefore, the work is focused on overcoming the limitations of the current mechanism by proposing an effective DL strategy called the Faster-PPENet. We have utilized the Faster-RCNN approach with a modified ResNet101 base to detect and classify various PPE. We have presented a robust feature extractor base by altering the conventional ResNet model.

Specifically, we have replaced the activation approach of the ResNet approach by utilizing the Swish activation technique as a substitute for the ReLU function to extract a more effective set of sample features which are later recognized by the 2-phase locator of the Faster-RCNN model. The model is evaluated on two complex databases to show the effectiveness of our approach. Further, a real-world data sample is also utilized to show the generalization ability of our approach. In the context of our deep learning-based research aimed at detecting PPE, the term logistics takes center stage as we address the practical complexities of deploying our advanced DL model. Beyond the development of a state-of-the-art deep learning algorithm for PPE detection, we emphasize the logistics of data collection and labeling, which are essential for robust model training. Furthermore, our project tackles the logistical challenges surrounding the seamless integration of the DL system into real-world workplace environments, encompassing considerations such as deployment logistics, operational efficiency, and timely maintenance. By optimizing these logistical aspects, our endeavor aims to not only enhance workplace safety through automated PPE detection but also provide organizations with a blueprint for the efficient implementation of AI-driven solutions. The distinctive contributions of the proposed approach are as follows:

- An effective DL approach called the Faster-PPENet is proposed that can locate and classify numerous types of construction-site safety equipment.
- Improved classification of PPE due to the better visual sample characteristics computation ability of the proposed approach.
- The model is effective in dealing with several images' transformational changes due to its empowerment to deal with the model's over-fitting data.
- The better generalization ability of the suggested approach is due to its increased recall ability.
- A vast evaluation of the approach is performed on a huge data sample and also tested on a real-world PPE repository to show the robustness of the suggested technique.
- An application system capable of monitoring the workers in the real environment is designed which can assist the supervisors and managers to track PPE compliance over time.
- Our work contributes by addressing the critical logistical aspects of deploying the DL method for PPE detection, including data collection and real-world integration. We offer practical insights to streamline DL implementation in workplace safety.

The remaining manuscript follows the mentioned scheme: Section II comprises related work, Section III elaborates on the proposed solution, Section IV comprises evaluation results, and Section V contains the conclusion.

II. RELATED WORK

There are several methods presented for the detection and identification of PPE from images. In this section, we have studied and presented a review of PPE-related existing studies. We have categorized the existing works into four subsections, where a generic pipeline for searching the related studies is provided in Figure 1.

A. YOLO-VARIANTS

Wang et al. [4] presented the DL-based method i.e., YOLO v5x for PPE detection for real construction sites. The authors proposed eight different methods on the images for real-time PPE detection and reported the best detector to YOLO v5x. The methods are evaluated on the CHV dataset which contains the 1330 images having six categories i.e., vest, persons, and four types of helmets. The comparative analysis is presented with other 8 models and achieved the highest results with YOLO v5x in terms of mAP which is 86.55%. The method gives overall good results, however, degrades performance in case of small, block objects and misclassification. It's also hard to detect helmets in the presence of a blurring effect in the images. Kumar et al. [28] proposed the YOLO v4 approach for PPE identification from images for construction sites. Therefore, reducing fatalities and financial damage can be considerably aided by the identification of PPEs, fire, and injured or trapped people as well. A revolutionary way of fire detection is presented, along with PPEs to help with monitoring and evacuation duties. The YOLOv4 and YOLOv4-tiny techniques are used in this study to complete the identification task. The Darknet model was trained using a self-created dataset. The performance of this method is good as mAP is 76.86%. However, performance is degraded on small objects. The method performed very well on the small dataset which needs to be trained on the larger databases and also in real-world scenarios. Lee et al. [29] designed a YOLO-EfficientNet model that comprises a hat, helmet, and head in different classes. The model architecture consists of two deep-learning approaches namely, classification and object detection. The first step is object detection where the head of a person is detected using the YOLOv5x algorithm. The second step is classification, which uses the EfficientNet87 algorithm to determine if the detected head has a helmet, hat, or nothing. The CNN-based classification model is applied for easy computation training and good performance.

B. CNN-BASED METHODS

Wu et al. [30] employed the method for the detection of hardhats from images. Most current works examining the existence and correct usage of hardhats on construction sites use multi-stage data processing, which has constraints on adaptability and universality. A one-stage approach based on a CNN is developed to automatically detect the hardhat color and determine if construction workers are wearing them. This work creates a brand-new benchmark dataset for hardhat-wearing detection that is freely accessible and contains 3174 photos that represent varied on-site circumstances. Then, using the suggested reverse progressive attention, fea-

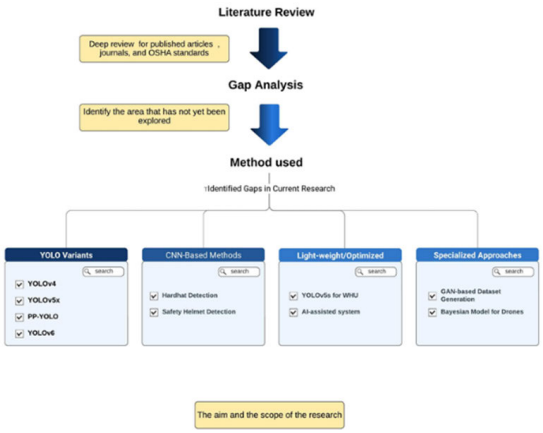


FIGURE 1. Generic pipeline to search the related existing studies.

tures from several layers with various scales are merged discriminately to create a new feature pyramid, which is fed into the SSD to forecast the final detection outcomes. This method achieved mAP=83.89%, which is a good performance however, the methods need to be improved for the identification of small hardhats. In [3], three models that are based on the YOLO are presented. These models may be used to instantly determine from images or videos if a worker is wearing a hard hat, vest, or both. These are detected in the first technique and then the ML approach is employed to check each worker is correctly donning a hat or vest. Secondly, CNN is utilized to instantaneously recognize specific workers, and approve PPE obedience. In the third method, the algorithm first isolates the employees in the input image before cropping and classifying them using VGG-16, ResNet-50, and Xception based on whether they are wearing PPE or not. The model achieved mAP= 85.6% in the case of the ResNet-50 approach. However, these methods are prone to occlusion, inadequate lighting, and blurriness. Shen et al. [31] employed a new procedure based on CNN to recognize the wearing of safety helmets. Initially, they detected the multiscale and small safety helmets from images. Secondly, the authors presented the transfer learning approach on DenseNet to correctly identify the safety helmets. A network model using two peer DenseNet systems is trained via cooperative distillation to significantly increase recognition precision. Numerous studies and trials reveal that the unique strategy has significant advantages over previous cutting-edge algorithms for identifying safety helmet use. However, the model is unable to identify employees who are facing away from the security camera.

C. LIGHTWEIGHT MODELS

Hayat and Morgado-Dias [32] proposed a DL-based approach named YOLOv5x to identify the safety helmets from images. The employed method is fast and automatic, which can process 45 frames per second. A Benchmark dataset was employed for experimentation, which contains 5000 images. The model is very efficient in detecting safety helmets, even in the case of small objects, However, performance is degraded in case of obstacles in front of helmets and entities identical to helmets. Similarly, Sadiq et al. [33] presented a helmet detection method from construction site images. The authors have presented the different techniques for object identification, in which YOLOv5 performed better than the other methods. After that, this method is customized using the YOLOv5 with fuzzy-based enhancement, which is a robust and efficient method for helmet detection. However, shows low performance in real-world scenarios under the presence of noise. Most of the work from the literature is related to YOLO, which has different versions, i.e., YOLOv3, YOLOv4, YOLOv5, and YOLOv5x, etc. Similarly, in the paper [34], a lightweight approach is utilized for helmet detection based on YOLOv3 named CSP-Ghost-ResNet. This is the lightweight model that reduces the complexity of the proposed approach and efficiently detects the safety helmets. However, accuracy is degraded when the focus is on reducing complexity. The SHWD is used for the experimentation of the proposed model. Another lightweight approach is presented in [35], which is based on YOLOv4. This model increases detection speed and accuracy however, unable to detect small objects. Similarly, Chen et al. [36] presented improved YOLOv4 for helmet detection from aerial images. This model is less complex as compared to YOLO v4 and achieved good results. YOLOv5 was utilized in many studies for helmet detection from construction site images. Overall, the performance is better and achieved comparable results. A lightweight face-assisted model that employs YOLOv5s to detect Welding Helmet Use (WHU) is recommended in [37]. It is implemented in two steps; the first step, the Ghost module uses YOLOv5s to optimize the extraction of feature parts such as backbone and neck to minimize model complexity. The second step involves reconstructing the neck of YOLOv5s based on Bi-directional Feature Pyramid network (Bi-FPN). In [38] a computer vision-based and AI-assisted embedded system to monitor human activities in critical environments to check on safety. The solution is based on modules responsible for particular tasks that can be easily configured by a user and be added to the system to provide more functionalities, e.g. measuring the social distances between people, estimating the number of pedestrians at a place, and detecting the kind of PPE worn. Two datasets are created to validate the framework.

D. SPECIALIZED MODELS

Jin et al. [39] proposed the DL-based technique for helmet detection from images. To support the system's capacity to distinguish between foreground and background, they employed K-means++ to enhance the size-matching unit of the preceding anchor box. Next, they integrated the Depth-wise Coordinate Attention method into the main network. The presented method performs well, however, it needs to be tested on other standard datasets. Worker safety is an essential

task at construction sites, for this purpose, workers must wear safety helmets. Automatic detection of safety helmets is needed because helmets can reduce injuries. In [40] a technique was proposed that employed the GAN-based approach for recognizing the safety equipment at the construction sites. The approach performs well for PPE identification, however, at the charge of an enhanced computing burden. According to the presented studies, most research is based only on Helmet detection, in which some of the papers are trying to detect other safety equipment in construction sites, i.e., Gloves, Glasses, vests, etc.

A framework that can detect objects on construction sites under different weather conditions is proposed in [41]. It employs generative adversarial networks (GAN) to form a dataset of images of a construction site in different weather conditions. The study aims to modify deep learning models to fit construction site conditions as well as develop models that detect objects in different weather conditions and enhance safety. It also develops an instance segmentation model that can be used in real construction sites. The use of drones has become widespread recently for various applications. To enhance the performance of drones, improvements in their functionality are crucial. Abbas et al. [42] developed a Bayesian model that provides the causal relationship between factors to influence the selection of drones based on their performance and, consequently, the probability of a drone performing in a conditioned salient criteria. The model was evaluated using analyses such as the Bayesian inference algorithm as well as other sensitivity analyses. Table 1 shows the review of related studies. These methods achieved good results, however, they need further improvements and also need real-time detection. So, there is a need for a model that can detect all safety equipment from construction sites in real-world scenarios and also detect small targets. The model also needs to produce alerts in case of missing safety measures.

TABLE 1. Review of related studies.

Method	Result	Dataset	Performance	Limitation
Wang et al. [4]	mAP=86.55%	CHV dataset	PPE detection has six classes	Low performance on small and block objects.
Kumar et al. [28]	mAP=76.86%	V6 dataset Self-created dataset	PPE identification	Unable to correctly detect small objects.
Wu et al. [30]	mAP=83.89%	GDUT-HWD	Hardhats identification	Performance needs to be improved for small hard hats.
Nath et al. [3]	mAP=85.6%	Pictor V3	Hardhats and Vest detection	Prone to occlusion, inadequate lighting, and blurriness.
Shen et al. [31]	Accuracy=94.47%	Data Fountain's dataset	Safety helmet Detection	The model is unable to identify employees who are facing away from the security camera.
Jin et al. [39]	Accuracy=95.9%	Proprietary dataset	Helmet detection	The method needs further improvements in terms of accuracy.
Hayat et al [32]	mAP=92.44%	MakeML dataset	Helmet detection	Low performance in hurdles in front of helmets, and things matching to helmets.
Sadiq et al. [33]	mAP=94.1%	MakeML, ROBOFLOW, SHWD	Helmet Detection	Low performance on real-world samples
Deng et al. [34]	mAP=90.9%	SHWD	Helmet Detection	Low accuracy
Chen et al. [35]	Accuracy=92.98%	SHWD	Helmet Detection	Unable to detect small targets
Wei Chen et al. [36]	mAP=91.03%	UAV aerial safety helmet	Helmet Detection	Need to improve in terms of accuracy

III. PROPOSED TECHNIQUE: Faster-PPENet

This research is based on the improved method for the detection of PPE from images. The presented method is based on the Faster-RCNN along with modified ResNet101 architecture, which introduces the Swish activation method. The modified Faster-RCNN approach is trained and tested on various datasets of PPE samples and given the name Faster-PPENet. Figure 2 describes our whole methodology in graphic form. The presented model proved that the Faster-RCNN with modified ResNet101 gives the best results for the recognition of PPE. To prepare the data for training the proposed Faster-PPENet model, several pre-processing steps were applied. All images were resized to 600 × 600 pixels, a dimension compatible with the input requirements of the ResNet101 backbone. This ensured uniformity across samples and compatibility with the model architecture. Annotations for object detection were initially created using the LabelImg tool, which allowed bounding box labels to be saved in the PASCAL VOC format.

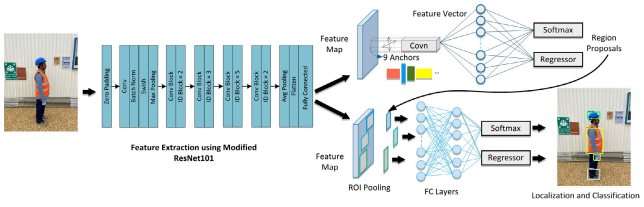


FIGURE 2. Faster-PPENet Framework.

A. FEATURE EXTRACTION USING Faster-RCNN

In this section, we presented our proposed technique in detail. For the recognition of PPE a Faster-RCNN [43] approach is adopted which is a DL-based method that has the ability

to detect objects from images. To make this approach more effective, we have employed modified ResNet101 for deep feature extraction from images. The reliable feature extraction method is very important in this field for effectual and accurate recognition of PPE. Because the larger set of features may lead to illogical behavior of the system and in the case of a small set there is a chance of missing the important keypoints. So, robust feature extraction can play an important and useful role in the identification of PPE from input images.

The Faster-RCNN technique uses convolutional filters which are proficient in analyzing the image structure to obtain the key points. The reason to choose this approach over the RCNN and Fast-RCNN is that it uses the RPN for feature selection. However, both methods, i.e., RCNN and Fast-RCNN rely on generic object proposals, which are based on hand-crafted features and a selective search approach. These methods face some problems and are time-consuming because of manual feature selection and also suffer from high computational complexity. Moreover, Faster-RCNN consists of two main phases: Regional Proposal Network (RPN) and Fast-RCNN. The first phase of RPN is fully convolutional to generate object proposals automatically which are then employed as input for the further steps. In the second phase, Fast-RCNN is utilized to refine the proposals generated from RPN, and the convolutional layer is common between both phases. So, input is delivered only once through the CNN to generate the proposals and then improve these object proposals.

B. MODIFIED ResNet101

ResNet101 is a CNN model that is part of the ResNet family. The ResNet [44] approach was proposed for addressing the vanishing gradient challenge in deep networks. ResNet101 has a total of 101 layers, which can be used for different tasks related to computer vision, i.e., object classification and recognition, etc. The layers in this network are structured into residual blocks, which further consist of multiple Conv layers having 33 blocks, and 29 of these squares are used in preceding blocks. In this work, we have modified this network by introducing the Swish function instead of the ReLU. The modified network diagram is illustrated in Figure 3. The Swish function can be calculated using Equation 1:

$$Swish(x) = x * sigmoid(x) \tag{1}$$

The swish function has some characteristic that gives promising results as compared to ReLU activation.

- The swish function is smooth as compared to ReLU which can be utilized during training to solve optimization and convergence issues.
- Swish solves the saturation problem which gives a nonzero outcome for positive and negative inputs.
- This activation function shows outperformance than ReLU for deep learning tasks related to computer vision.

In the field of computer vision and in real-world problems, there are two main challenges: the correct location

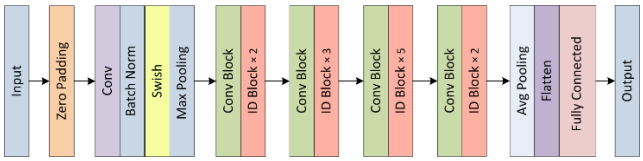


FIGURE 3. Modified ResNet101.

of multiple objects from the image and specifying the true class of detected objects. To detect and recognize the PPE from images, Faster-RCNN overcomes the above challenges by using the RPN module and Fast-RCNN. RPN utilizes the knowledge about objects' size, color, etc., to detect the class and its location, which gives a higher performance rate. Moreover, it reduces the overall computational cost and achieves good results. This approach contains the following main steps:

C. CONVOLUTION LAYERS

ResNet-101 consists of multiple stacked residual blocks, where each block contains convolutional layers and identity shortcuts. The architecture has 101 layers in total, hence its name ResNet-101. The stacked convolutional layers in the residual block are accountable for calculating features from the image. In ResNet-101, there are multiple convolutional layers (usually 3×3) in each residual block. These layers are followed by batch normalization and Swish function to present non-linearity. The extracted samples feature map is then passed among the RPN and related linked layers.

D. RPN

The primary purpose of the RPN module is to propose candidate areas in an image that are possible to encompass objects. These candidate regions are then employed as input to the succeeding object recognition pipeline. The RPN generates region proposals by sliding a small window (usually 3×3) over the convolutional feature map of the input image. For each window location, the RPN predicts whether an object is present in the window and refines the window to more accurately fit the object.

E. ANCHOR BOXES

The RPN uses predefined anchor boxes of various sizes and aspect ratios. These anchor boxes are centered at each sliding window position on the feature map. The RPN predicts whether each anchor box contains an object and, if so, the refinement offsets needed to match the object's position.

F. CLASSIFICATION NETWORK

The classification sub-network is responsible for predicting whether each anchor box contains an object or is just a background region. It predicts the objectness score for each anchor box, indicating the probability of the box encompassing an object. The classification network uses a cross-entropy loss to train and optimize the objectness scores.

G. REGRESSION NETWORK

The regression sub-network predicts the refinement offsets for the anchor boxes that are classified as containing objects. These offsets are used to adjust the anchor boxes' positions and sizes to better fit the actual objects' locations and shapes. The regression sub-network uses smooth L1 loss (Huber loss) to train and optimize the refinement offsets.

H. NON-MAXIMUM SUPPRESSION (NMS)

After obtaining the objectness scores and refined bounding box coordinates for all anchor boxes, non-maximum suppression is applied to remove duplicate and highly overlapping proposals. NMS confirms that only the maximum assured and non-overlapping proposals are reserved, decreasing redundancy and computational load in subsequent stages of object detection.

IV. RESULTS

This part of the manuscript holds information about the data samples used to train and test the proposed approach, along with the evaluation metrics. Further, an in-depth investigation of the introduced work involving numerous experiments is also accomplished to prove the effectiveness of our approach to PPE recognition.

A. DATASET

To check the performance of our method, we have adopted two standard datasets, i.e., CHV [4] and SHD [45]. The CHV dataset comprises a total of 1330 samples nominated from the ten thousand images with complex selection criteria to include the PPE samples with complex background settings. The dataset comprises a total of six classes, namely person, vest, and helmet with blue, white, yellow, and red colors. The SHD dataset comprises a total of 5,000 images with three classes called person, helmet, and head, respectively. Both datasets are challenging and comprise samples with diverse capturing conditions like distant objects, light, and color variations.

B. PARAMETERS

To numerically elaborate the performance results of the presented work, various standard metrics are chosen like intersection over union (IoU), mean Average Precision (mAP), accuracy, precision, F1-score, and recall. The model categorization accuracy is computed by using Equation 2.

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN} \tag{2}$$

Equation 3 exhibits the description of the mAP score, where the AP specifies the average precision computed for all groups, and $t$denotes the processed sample. While T designates the total samples.

$$mAP := \sum_{i=1}^{T} AP(t_i)/T \tag{3}$$

Figure 4 denotes the graphic form of precision, recall, and IOU, respectively.

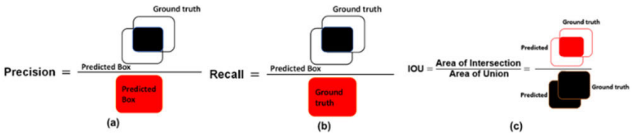


FIGURE 4. Graphic depiction of (a) Precision, (b) Recall, and (c) IOU measures.

C. EVALUATION OF THE PROPOSED APPROACH

To discuss the localization and classification results of the Faster-PPENet, we have performed various experiments in this section. Initially, the detection performance of the proposed approach is evaluated; then, we discuss the class-wise classification values of the suggested strategy. Finally, the attained results on the entire employed databases are elaborated to show the effectiveness of our approach to recognizing the PPE objects from the given samples. A detailed analysis can be found in the subsequent sections.

1) LOCALIZATION RESULTS OF THE FASTER-PPENET

The major aspect of any recognition system is assessed by its ability to locate the area of interest. To investigate this, we have carried out an evaluation here in which we have taken the samples from both datasets, named the CHV and SHD, and tested them via utilizing the trained model. The attained localized results showing the recognition ability of the suggested strategy are reported in Figure 5. The images in Figure 5 depict that the introduced framework can locate various types of PPE objects with a high recognition ability. Moreover, it is quite visible that the proposed work can tackle the complex sample background conditions and identify the PPE objects exactly from the investigated images. Further, to quantitatively elaborate on the resultant values, we have computed the mAP, and IOU scores for both employed databases. The main reason for computing the mentioned evaluation values is that the mAP, and IOU are considered the standards to numerically report the recognition ability of an object detection model. Specifically, for the CHV dataset, we have attained a mAP value of 0.8845 along with an IOU score of 0.8812. While for the SHD dataset, we have acquired the mAP and IOU values of 0.9221 and 0.9118, respectively. The reported results show the robustness of the proposed work in recognizing the various PPE objects.

D. CLASS-WISE VALUES DISCUSSION

In this section, the class-wise results values for both employed datasets are deliberated as discussing class-wise results helps to communicate the model's strengths and weaknesses effectively. It allows domain experts to understand which classes the model excels at and where it struggles.

Initially, the category-wise values attained for the CHV dataset are reported with the computation of different



FIGURE 5. Localization results of the Faster-PPENet.

standard metrics like precision, recall, accuracy, F1, and error rate. First, precision and recall values are computed for all six classes of the CHV dataset, and the acquired results are shown in Figure 6. The scores visible in Figure 6 depict that the proposed approach is effective in recognizing all classes of CHV data samples. More clearly, in the aspect of precision, the Faster-PPENet approach has reported values of 91.18%, 88.21%, 93.13%, 84.48%, 90.15%, and 94.08% for the person, vest, blue, red, white, and yellow helmet groups. Then, in the perspective of recall, the Faster-PPENet method has acquired scores of 86.66%, 86.62%, 87.09%, 95.16%, 92.22%, and 92.28% for all the above-mentioned six classes, respectively.

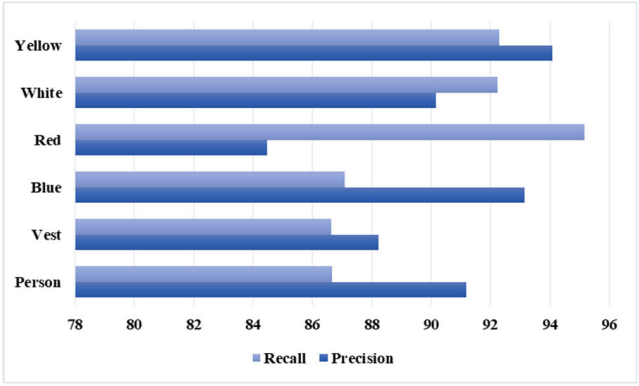


FIGURE 6. Faster-PPENet precision, and recall results for the CHV data sample in aspect of all six classes.

Next, the value results for the CHV dataset are presented in the form of F1, along with the error rate for all six classes, and the obtained scores are mentioned in Figure 7. It is evident from the scores given in Figure 7 that our approach has exhibited better F1 values for all six groups with accepted values of error rates. Clearly, we have acquired F1 scores of 88.86%, 87.41%, 90.01%, 89.50%, 91.17%, and 93.17% for the person, vest, blue, red, white, and yellow helmet groups. Further, for the CHV dataset, the Faster-PPENet approach has shown the smallest and highest error scores of 6.83% and 12.59% for the yellow helmet and vest classes, respectively. The reported scores show that our approach can recognize all categories with high scores due to its better generalization power.

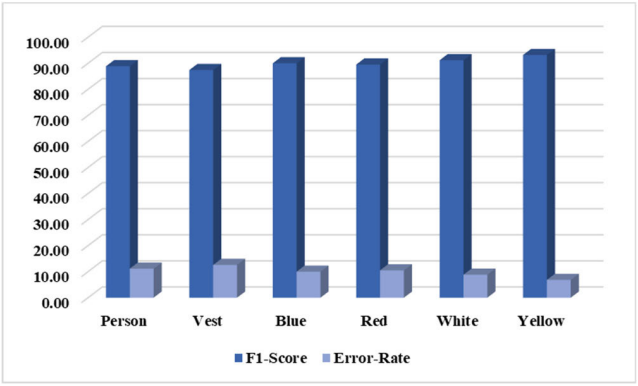


FIGURE 7. Faster-PPENet F1-score and error rates result for the CHV data sample in the aspect of all six classes.

Next, results in terms of accuracy are discussed for all classes of the CHV data sample, as computing accuracy is a critical aspect of image classification because it measures the performance and reliability of the classification model. This metric assists in quantifying the correctness of the model's predictions by comparing them to the ground truth labels (the correct class labels) of the images. The attained scores for the CHV dataset are given in Figure 8 from where it can be judged that the Faster-PPENet model attains effective values for all classes of the employed database. Clearly, the lowest accuracy is reported for the red helmet class with a score of 84.51%, while the highest value is acquired for the yellow helmet class with a value of 94.14%, which indicates the effectiveness of our work.

After the CHV dataset, the classification values for the SHD data sample are discussed. First, values are shown in terms of precision and recall measures, and attained analysis is given in Figure 9. The scores in Figure 9 clearly show that the proposed approach is proficient in differentiating all three classes of the SHD dataset with high recognition ability. Clearly, for the person, helmet, and head classes of the SHD dataset, the Faster-PPENet approach has attained precision scores of 94.42%, 94.72%, and 94.54%, which are 92.87%, 92.92%, and 92.64% in the aspect of recall measure. Next, the classification values in the form of F1-score and error

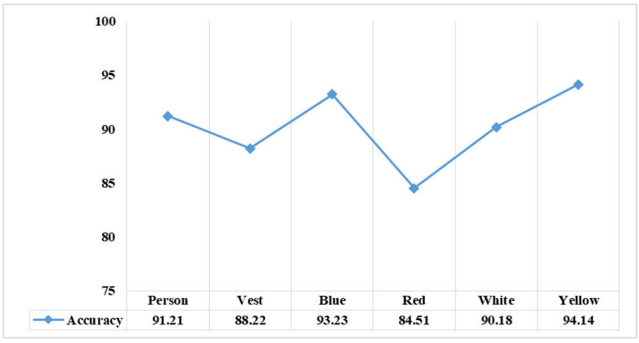


FIGURE 8. Faster-PPENet accuracy result for the CHV data sample in the aspect of all six classes.

rates are presented in Figure 10, where it is evident that the proposed approach has attained effective classification results by showing small values for error rates. Clearly, we have attained an average error rate of 6.32%, which indicates the robustness of our approach.

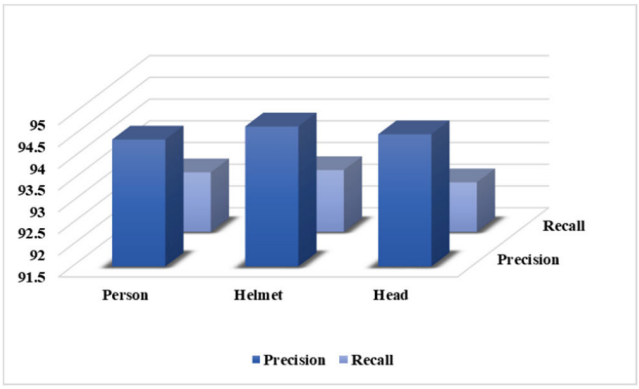


FIGURE 9. Faster-PPENet precision, and recall results for the SHD data sample in the aspect of all three classes.

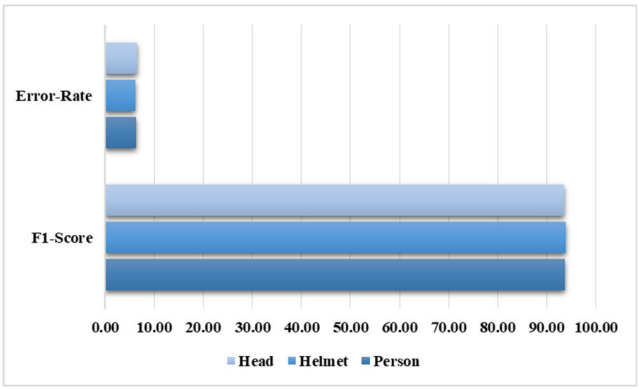


FIGURE 10. Faster-PPENet F1-score and error rates result for the SHD data sample in the aspect of all three classes.

Finally, the classification values of the Faster-PPENet approach are elaborated in the aspect of the accuracy metric for all three classes of the SHD data sample. The obtained accuracy numbers for the person, helmet, and head classes are shown in Figure 11. The proposed approach performs well for all three classes of the SHD dataset with better accuracy scores. Clearly, we have attained the accuracy values of 93.88%, 94%, and 94.01% for the person, helmet, and head groups, which proves the effectiveness of our approach.

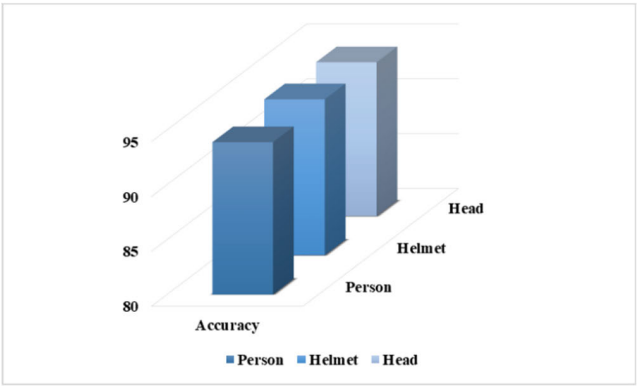


FIGURE 11. Faster-PPENet accuracy result for the SHD data sample in the aspect of all three classes.

After accomplishing an extensive investigation of the PPE object recognition ability of the proposed approach, we can say that our suggested framework is effective for this domain. The major cause for this enhanced recognition ability is the better visual information-capturing capacity of the Faster-PPENet model, which enhances the recall ability of the approach and increases the classification results.

1) ENTIRE DATASET

In this part, we have designed an experiment to discuss the performance of the proposed approach on the entire samples of the CHV and SHD datasets. In summary, discussing classification results on the entire image dataset is essential for understanding the model's performance across various classes, which assists in revealing the model's strengths and weaknesses and identifying the potential areas for further improvement. The performance comparison of the Faster-PPENet approach for both data samples in the aspect of standard evaluation parameters is provided in Figure 12. The values show that the suggested network can recognize the classes of both data samples with a high recall rate. Particularly, in the aspect of accuracy, we have acquired scores of 90.25% and 94.03% for the CHV and SHD datasets. The empowerment of the approach to propagate a reliable sample characteristic set under altering background conditions with high recall ability allows the Faster-PPENet model to recognize all classes of the employed datasets and is well suited to accomplish the multi-class classification problem.

E. COMPARISON WITH BASE MODELS

This part comprises the model evaluation for both databases named the CHV and SHD datasets with numerous other

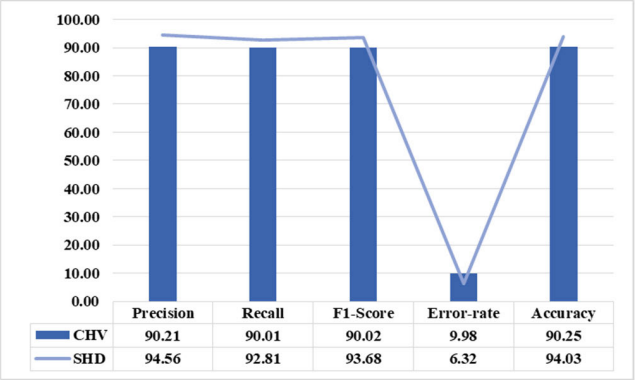


FIGURE 12. Faster-PPENet performance results on both CHV and SHD datasets.

object detection techniques to compare the PPE recognition capability of our approach against them.

Initially, we reported the results for the CHV data repository. For this reason, we have selected various object detection approaches like RCNN, Fast-RCNN, conventional Faster-RCNN, and several variants of the YOLO framework. We have compared the result values in the aspect of mAP metric as it is considered the standard evaluation measure in object localization. The attained comparison of the Faster-PPENet with the base models is shown in Table 2, from where it can be witnessed that our approach performs better than all other base approaches. Specifically, the lowest values are reported by the RCNN approach with a mAP of 0.8143, while the second lowest value is attained by the YOLOV3-5-layered approach with a mAP of 0.8199. The conventional Faster-RCNN approach obtained a mAP score of 0.8605, while a comparable result is shown by the YOLOV5-large with 0.8614. In comparison, the proposed Faster-PPENet approach acquires the highest mAP score of 0.8845. More descriptively, the base approaches have attained an average mAP value of 0.8384, which is 0.8845 for the introduced work. So, for the CHV dataset, we have reported a performance enhancement of 4.61%. The main cause for this better recognition performance power of the suggested approach is that the RCNN and Fast-RCNN models utilize a hand-coded feature engineering technique to compute the sample characteristics, which lack the in-depth visual details of a given image. The conventional Faster-RCNN approach does not support the propagation of negative scores during the feature extraction stage, which decreases the recall ability of this approach. Moreover, the variants of the YOLO frameworks cannot locate small objects, which in turn decreases the PPE recognition ability of these approaches. In contrast, the proposed approach has a better deal with the issues of existing approaches by proposing a more effective feature engineering base and can easily locate the equipment of all sizes under complex samples transformational alterations and enhances the PPE detection and classification power of the proposed approach.

TABLE 2. Comparison of the Faster-PPENet with the base models for the CHV database.

Model	mAP
RCNN [46]	0.8143
Fast-RCNN [47]	0.8227
Conventional Faster-RCNN [48]	0.8605
YOLOV3-5-layered [49]	0.8199
YOLOV4-416-layered [50]	0.8416
YOLOV4-608-layered [51]	0.8484
YOLOV5-large [52]	0.8614
Proposed	0.8845

Next, the results are discussed for the SHD dataset against numerous base object detection techniques like SSD, YOLO-V3, EfficientDet-D0, and CenterNet models. The obtained comparison with all selected models in the aspect of mAP scores is given in Table 3, from where it can be seen that our approach performs better than all other base approaches with the highest mAP value of 0.9221. Moreover, the lowest value is reported by the SSD model with a mAP score of 0.796. The CenterNet and conventional Faster-RCNN models show better helmet detection performance with mAP scores of 0.880 and 0.899, respectively. However, our approach outperforms the existing approaches with a performance gain of 8.33% in comparison to all other approaches. The Faster-PPENet approach performs better than other approaches due to its effective keypoints calculation capability under altering background conditions and better recall ability. So, based on the analysis discussed in this section, we can say that our approach is more proficient in recognizing the PPE objects than the peer approaches.

TABLE 3. Comparison of the Faster-PPENet with the base models for the SHD database.

Model	mAP
SSD [53]	0.7960
YOLO-V3 [49]	0.8040
EfficientDet-D0 [54]	0.8650
CenterNet [55]	0.8800
Conventional Faster-RCNN [48]	0.8990
Proposed	**0.9321**

F. COMPARISON WITH THE STATE-OF-THE-ART

This part shows the investigation of the proposed approach in comparison to other latest techniques for both datasets named the CHV and SHD, respectively.

Initially, the result comparison for the CHV dataset is performed with numerous latest approaches in the aspect of both class-wise and overall mAP values, and the attained evaluation is presented in Table 4. The scores in Table 4 clearly show that the proposed work is more robust for PPE recognition as compared to all other latest approaches both in the aspect of the individual classes and on the entire data sample. In [3], an object detection approach named the YOLO-V3

was proposed to localize and categorize the various PPE samples and showed accumulative mAP values of 0.7230, while in [30], a single-stage object locator approach was introduced to locate the presence of PPE in the input samples and attained a mAP score of 0.8179. Moreover in [4], a DL framework called the YOLO-V5x was presented for recognizing the PPE-wearing samples of humans with a mAP of 0.8655. Comparatively, the proposed approach, called the Faster-PPENet, performs better for PPE recognition with a mAP score of 0.8815 over the CHV data sample which indicates the effectiveness of our approach. Clearly, in the perspective of person class, the competitor works have reported an average mAP number of 0.8377, which is 0.8488 for our work, and we have acquired a performance enhancement of 1.11%. For the Vest class, the peer works have shown an average mAP of 0.8322, which is 0.8507 for our work, with a performance improvement of 1.85%. For the Blue, red, white, and yellow helmet classes, the peer works have reported average mAP scores of 0.7906, 0.8025, 0.823, and 0.8495, while, comparatively, the Faster-PPENet approach has shown the average mAP scores of 0.8226, 0.9331, 0.9187, and 0.9348, respectively. So, for the blue, red, white, and yellow helmet classes, we have presented performance gains of 3.20%, 13.06%, 9.57%, and 8.53%, respectively, which indicate the robustness of our approach. Moreover, in the entire data sample, the selected approaches have reported an average mAP of 0.8021, which is 0.8845 for our case, so, we have attained an overall performance enhancement of 8.26%.

TABLE 4. Comparison of the Faster-PPENet with the CHV dataset.

Method	Class-wise mAP						Overall mAP
	Person	Vest	Blue	Red	White	Yellow	
YOLO v3 [3]	-	0.8496	-	-	-	-	0.7230
Pelee-RPA [30]	-	-	0.7735	0.6858	0.7464	0.7848	0.8179
YOLO v5x [4]	0.8377	0.8147	0.8076	0.9191	0.8996	0.9141	0.8655
Proposed	0.8488	0.8507	0.8226	0.9331	0.9187	0.9348	0.8845

Next, the performance comparison for the second dataset called the SHD is given in Table 5 for several latest approaches as given in [56]. The method in [57] applied a DL approach with a cross-layer attention strategy with a mAP value of 0.8320. The method in [58] proposed an object detection approach called the CSYOLOV3 and attained a mAP of 0.871. Moreover, Tan et al. [54] introduced a DL framework called the EfficientDet and evaluated the SHD data sample with a mAP of 0.865. Further, an approach called the RBFPDet was introduced in [56] for recognizing the PPE samples and showed an average mAP of 0.917. Comparatively, the proposed approach has attained the highest mAP score with a value of 0.9221. Clearly, the competitor methods have shown an average mAP of 0.8713, which is 0.922 for the Faster-PPENEt. Therefore, we have attained a performance gain of 6.09%.

TABLE 5. Comparison of the Faster-PPENet with the SHD dataset.

Method	mAP
Cross-layer Attention Mechanism [57]	0.8320
CSYOLOv3 [58]	0.8710
EfficientDet [54]	0.8650
RBFPDet [56]	0.9170
Proposed	**0.9321**

Based on the result comparisons given in Table 4, and Table 5 for the CHV and SHD datasets, it is quite evident that our approach is more proficient in locating and recognizing the PPE from the given samples. The major characteristic of the Faster-PPENet approach is the effective features calculation empowerment of the suggested strategy. The inclusion of the Swish activation approach in the feature extractor of the Faster-PPENet approach results in computing a more distinctive set of sample features. The custom feature extractor permits the proposed approach to enable the flow of negative scores during the information engineering process and allows the resultant values to decline even for the high input scores which results in improving the data storage capability of the Faster-PPENet approach. Such structure of the suggested approach allows it to compute a robust key point set of underlying samples comparatively to other peer approaches and results in improved PPE recognition performance.

G. GENERALIZATION ABILITY TESTING

To investigate the generalization capability of the Faster-PPENet, an experiment is accomplished in this part of the manuscript. For this reason, we have gathered a real-world dataset under diverse environmental conditions. We have gathered a total of 1000 samples.

The model is tested on this custom dataset to check whether it can locate the PPE objects from real-world scenarios. We have computed the performance of the proposed approach for the real-world sample with the assistance of numerous standard measures. The visual localized samples are provided in Figure 13, indicating that the proposed work is proficient in recognizing PPE objects from real scenarios with a high recall power. Particularly, we have attained a mAP score of 0.901 on this sample, which proves the generalization power of the suggested work. To further numerically discuss the classification behavior of the proposed approach, the computed accuracy, precision, F1, and recall scores are provided in Figure 13. Clearly, we have attained the computed accuracy, precision, F1, and recall scores of 91%, 91.10%, 89.96%, and 90.52%. Both the numeric and pictorial evaluation of the approach prove that the Faster-PPENet can

recognize all types of PPE objects due to its high recall and generalization ability. Further, the model is effective in dealing with several images' transformational changes due to its empowerment to deal with the model's over-fitting data.

H. APPLICATION DESIGN FOR PPE

Further, we have designed an application that can monitor if the persons are wearing PPE or not. Such application can be beneficial, especially in contexts where safety, health, and compliance with regulations are crucial. In various industries such as construction, manufacturing, healthcare, and hazardous material handling, the use of PPE is essential to protect workers from potential risks and hazards. By having an application that can detect whether individuals are wearing the necessary PPE, employers can promote a safer working environment and reduce the likelihood of accidents and injuries. Many industries have strict regulations and guidelines that mandate the use of specific PPE in certain work environments. By having a PPE detection application, organizations can ensure that they are adhering to these regulations, avoiding potential legal and financial repercussions. Our designed PPE detection application is equipped with monitoring and reporting features. These tools allow supervisors and managers to track PPE compliance over time, identify patterns, and address any issues or shortcomings promptly. A view of the developed application is provided in Figure 14.



FIGURE 13. Localized results on the proprietary dataset.

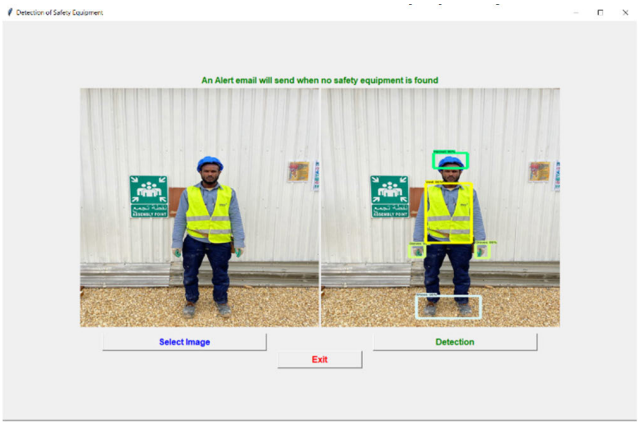


FIGURE 14. Application view of the Faster-PPENet.

V. CONCLUSION

The work has proposed a DL-based approach named Faster-PPENet to monitor whether the workers at working sites are wearing PPE objects or not. Clearly, an improved Faster-RCNN model is suggested by introducing an effective feature extractor. We have altered the existing architecture of the ResNet-101 model by replacing its activation method to swish from ReLu and utilized it as the base network of the presented work. The model is evaluated on two complex databases named the CHV and SHD to show the effectiveness of our approach. Further, to show the generalization ability of the proposed work, a real-world data sample is also collected and evaluated on the Faster-PPENet. For the CHV and SHD data samples, we have attained mAP scores of 0.8845, 0.9321, along with accuracies of 90.25%, and 94.03%, while for the real-world sample, we have attained the mAP, and accuracy values of 0.901, 91% of which is clearly proving the robustness of our approach. Moreover, we have designed an application system capable of monitoring the workers in the real environment, which can assist the supervisors and managers in tracking PPE compliance over time. Based on extensive analysis both in the aspect of numeric scores and visual depiction, we can conclude that the proposed approach is effective in PPE recognition. Despite promising results, the proposed approach has limitations, including reduced performance under extreme lighting or occlusion and high computational demands that limit use on edge devices. It also focuses only on select PPE items like helmets and vests, excluding others such as gloves and goggles. Future work will explore lightweight models (e.g., YOLOv8, MobileNet-SSD), attention mechanisms, and broader datasets to enhance real-time performance, generalization, and PPE category coverage, guiding further research in practical safety monitoring systems.

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